PVC COOLING SYSTEM

Assembly Instruction

IMPORTANT : It is very important that the step in this manual are followed EXACTLY. Failure to do so may result in difficuit assembly and/or part shortages.

Contents

1. Safety Instruction 3
2. Tools Required 3
3. Required Wall Opening 4
4. Installing the Support Bracket 5
5. Installing the Top Bracket 5
6. Installing the Water Gutter L3m 6
7. Installing the Float Gutter 7
8. Installing the Top Profile 8
9. Installing the End Cover 9
10. Installing the Cooling Pad ..... 10 - 11
11. Cool Pad Extension Bracket Assembly ..... 12 - 13
12. Part List Distribution 14
13. Water Pump Specification 15
14. Maintenance 16

Disclamier

KEHE reserves the right to make alterations to specifications, quantities, dimensions etc . for production or other reasons subsequent to publication.

● The information obtained herein has been prepared by qualified experts within KEHE While we believe the information is accurate and complete, we make no warranty or representation for any particular purpose.

● The information is offered in good faith and with the under-standing that any use of the units or accessories in breach of the directions and warnings in this document is at the sole discretion and risk of the user.

Introduction

● PVC Cooling System is PVC pad gutter system.
● It can be used for all poultry animal husbandry and greenhouse.
● Its a modular system which can be applied to every dimension(length or height).
● It including BOX 1 (6MT SYSTEM) and BOX2 (6MT SYSTEM).

Delivery Check

● Please inspect the external damage before opening the boxes.
● If found contact your forwarding agent before opening.
● If safe, please open the boxes and compare the quantities of products in each box with the parts list.
● Please note that the boxes includes small and numerous parts that can be lost easily.
● Please do not open the boxes if not be used and keep it in a closed and safe,dry place.

Notes

KEHE do not guarantee to inform users about changes or to distribute new manuals. All rights reserved. No part of this manual can be copied or published without the written authorization of KEHE . The contents of this manual can be changed without any notice .

1. Safety Instructions

Install and Operate Electrical Equipment Properly



Rread and understand all servide manual instruction before operating or servicing this Cool Cell System.



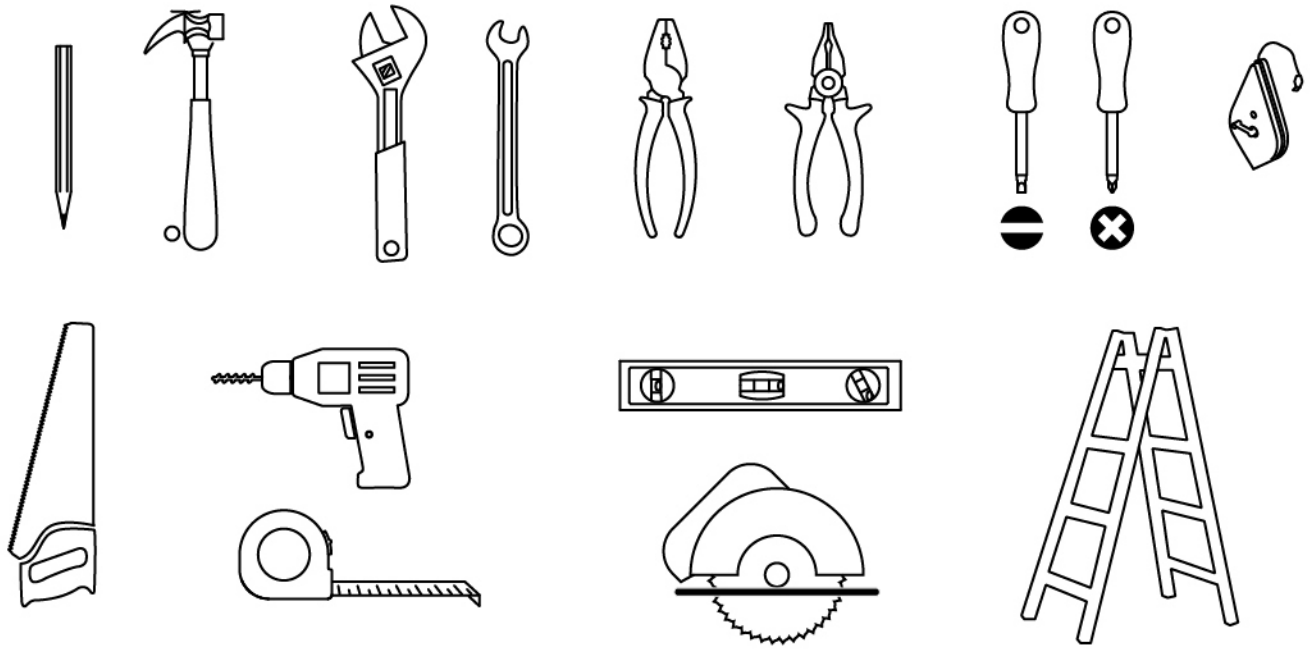
Electrical controls must be installed by a qualified electrician and must meet the standards set by applicable local codes.
Lock-out power source before making adjustments, cleaning, or maintaining equipment.
Bare cables can cause an electric shock when touching and can result in a short circuit.



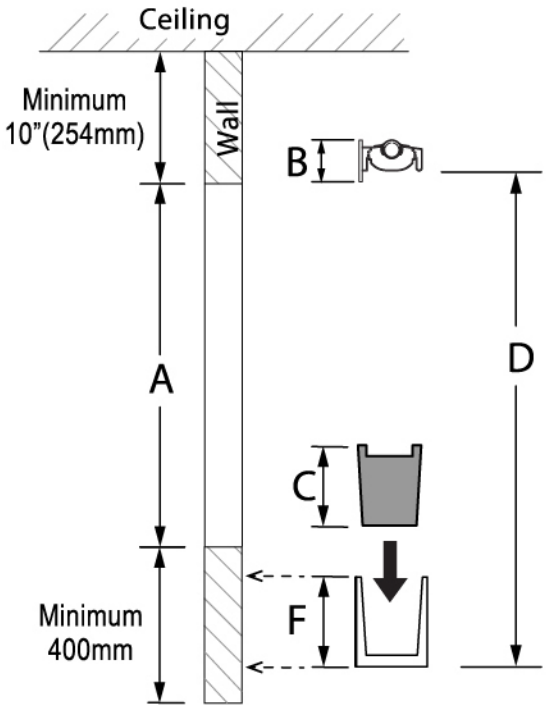
Warning !

DO NOT smoke, weld or grind in the cool cell work area.
No open fore hot air blower, gas burners in the cool cell work area.

2. Tools Required

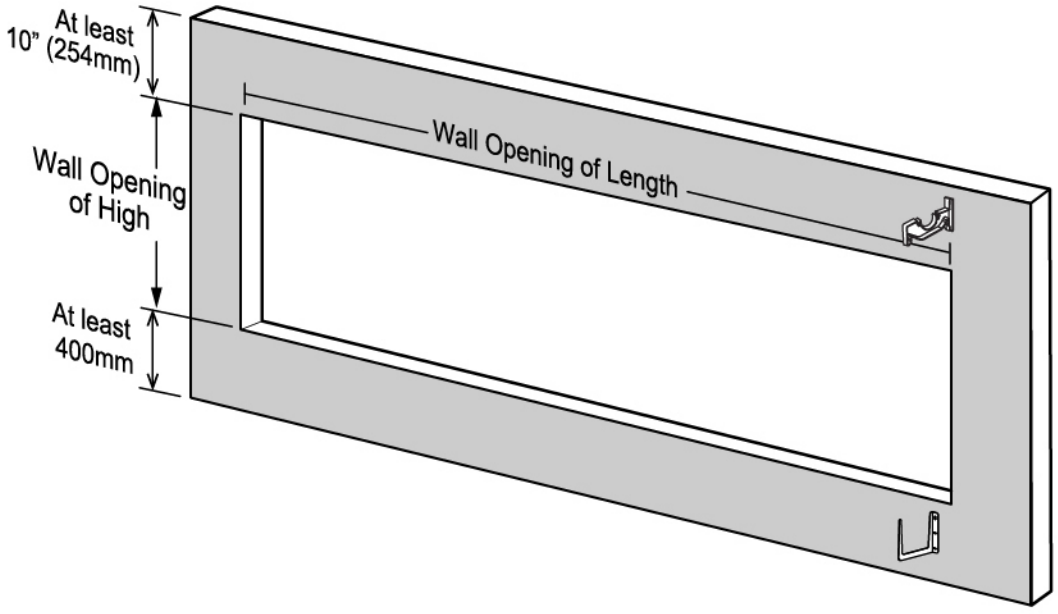


Cross Section



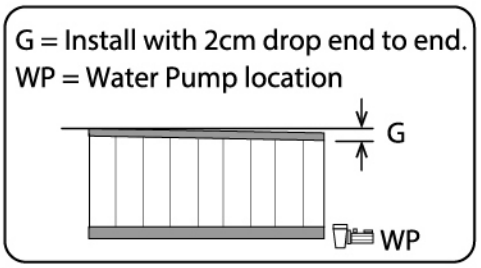
A	A = Wall opening A = F - 100mm (4 inches)
B	B = Top Bracket Height = 135mm
C	C = Water Gutter Height = 260mm Length = 3 meter
D	Distance between Top Bracket to bottom level of Support Bracket
E	Height of Cooling Pad 1530mm (5ft) 1830mm (6ft)
F	Support Bracket Height = 290 mm

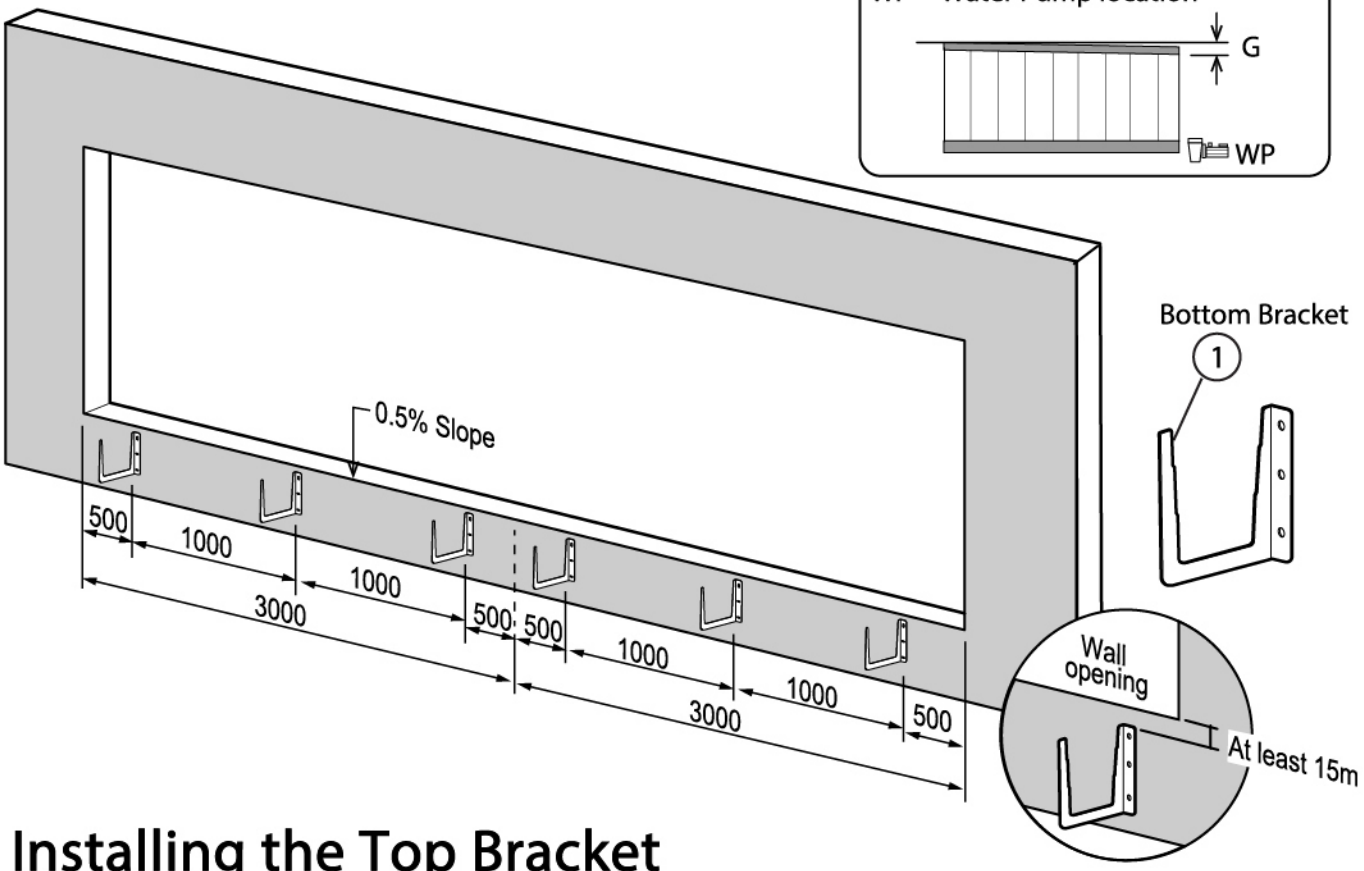
3. Required Wall Opening



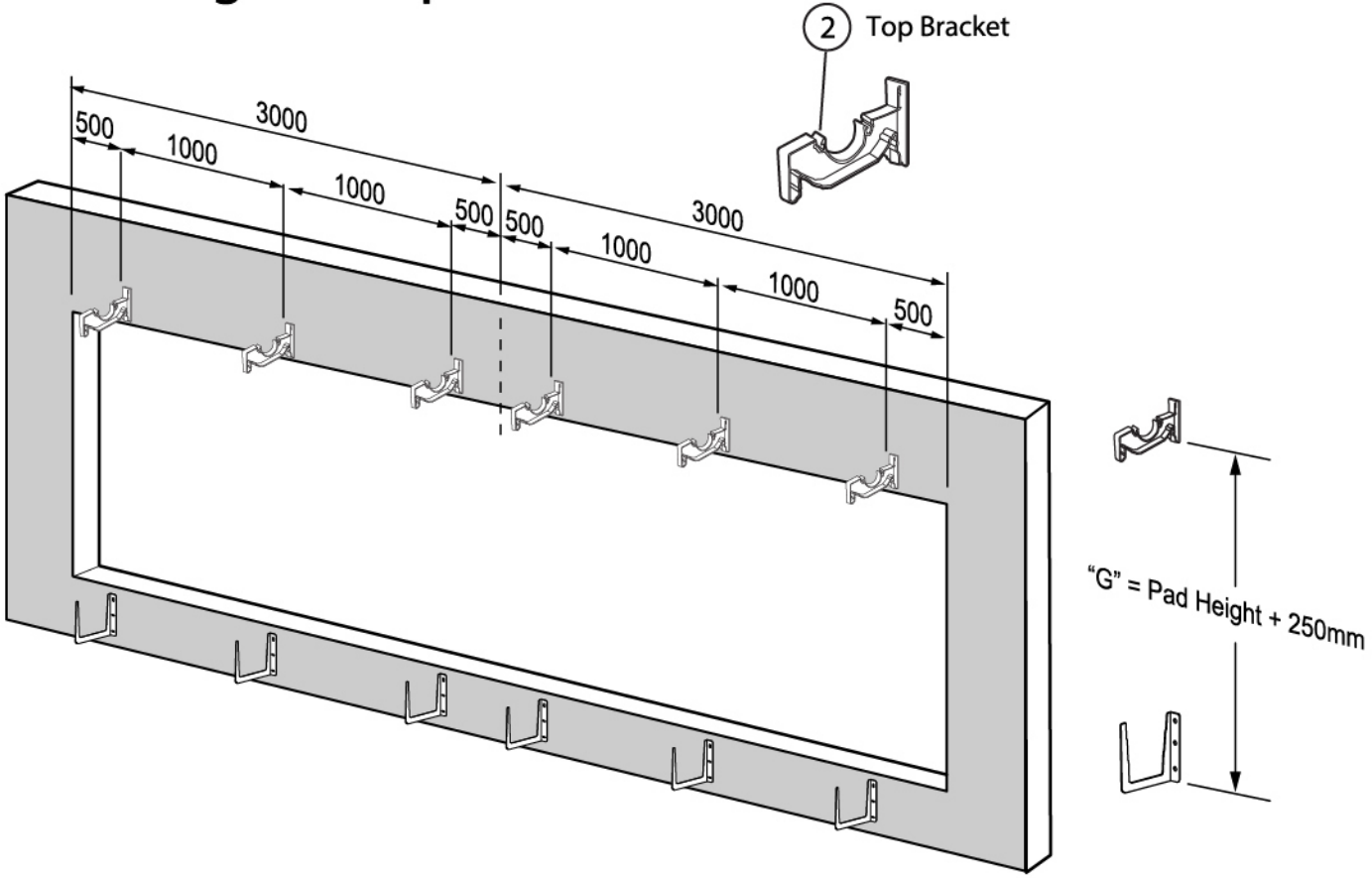
Length Opening	Height Opening
= Cooling Pad length - 100mm Example : 6000mm - 100mm = 5900mm	= Cooling Pad high - 100mm Example : 1530mm - 100mm = 1430mm 1830mm - 100mm = 1730mm

4. Installing the Support Bracket





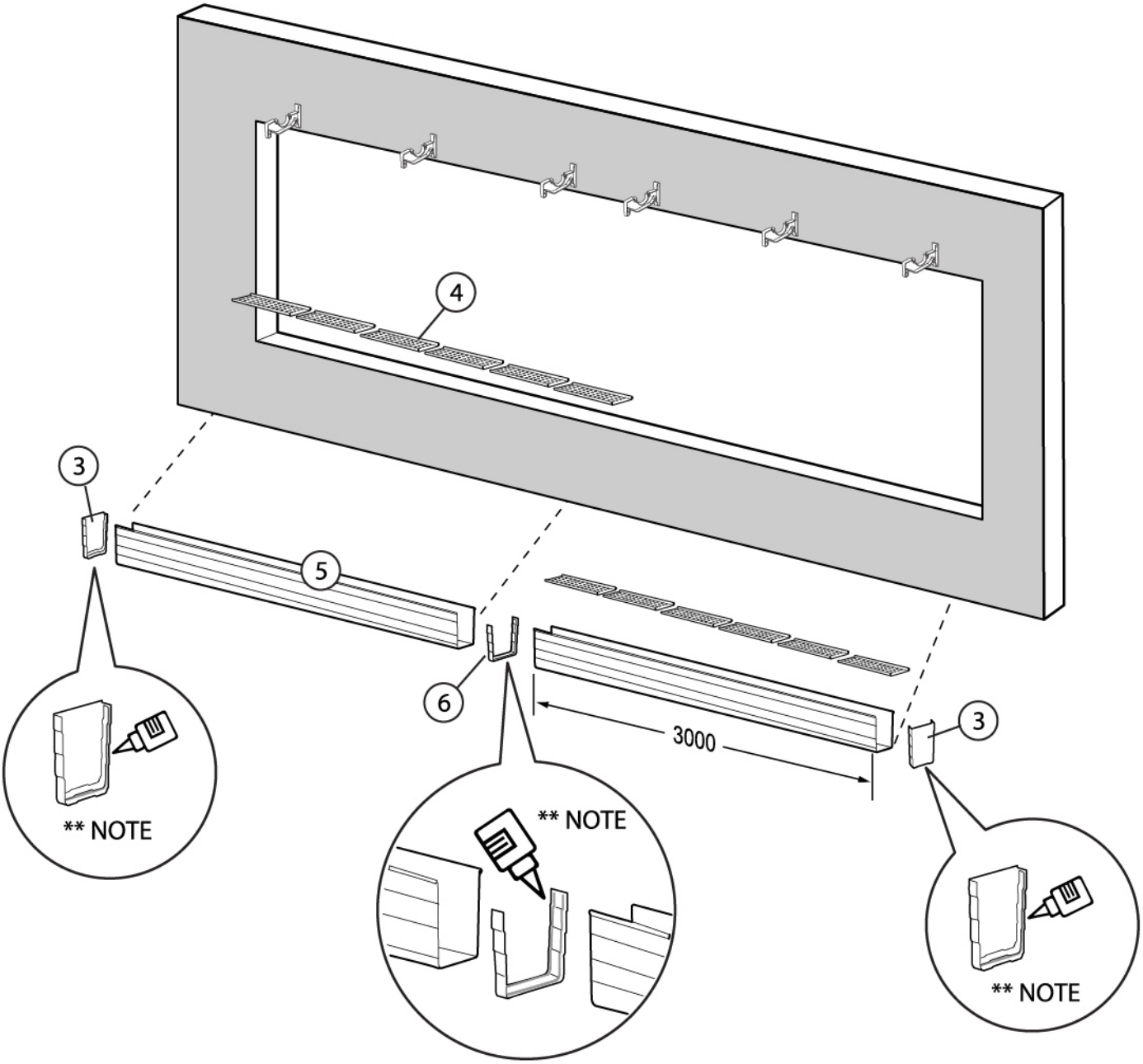
5. Installing the Top Bracket



Distance "G" :
Distance between Top Bracket to the bottom level of the Support Bracket is cooling pad height +250 mm.
Example: 1750mm for a 1530 mm (5FT) height pad or 2050 mm for a 1830 mm (6FT) height pad

6. Installing the Water Gutter, L3m

Ref #	Description
1	Support Bracket
2	Top Bracket
3	End Cover for Gutter
4	Pad Support, L500mm
5	Water Gutter, L3000mm
6	Coupling for Water Gutter



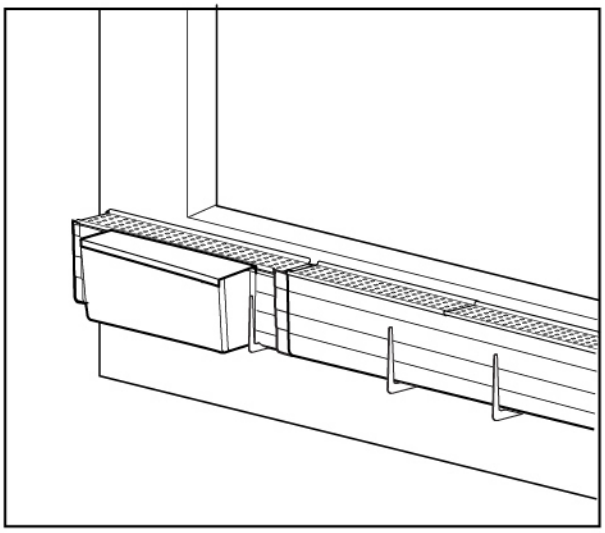
** NOTE : Fill the groove of coupling with PVC Oil Bottle With Heavy Duty Solvent Cement

NEVER use solvent cement and screw tight outside the gutter.

7. Installing the Float Gutter

The Float Gutter , L500mm supply can be installed in 2 different variations.

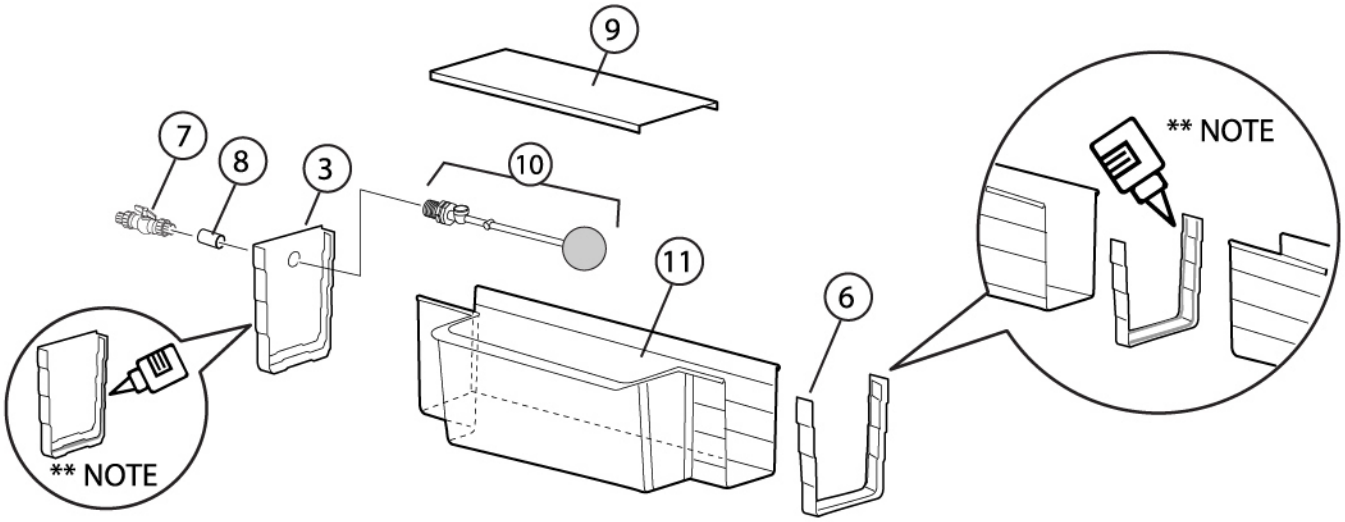
OPTION 1 : End Mount Method



Install the Float Gutter to the end of cooling system.

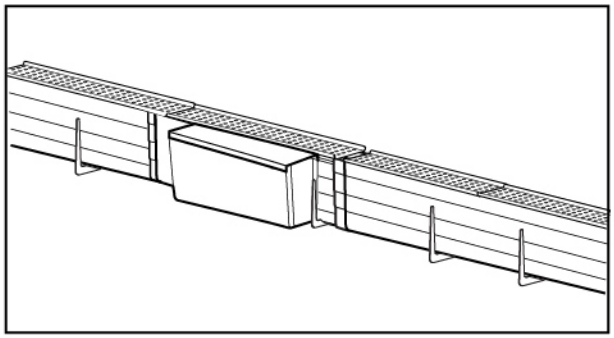
Ref #	Description
3	End Cover for Gutter
6	Coupling for Water Gutter
7	25mm Ball Valve with /Coupling
8	25mm Pipe
9	Float Cover
10	Float
11	Float Gutter Tank W/ 2 Coupling

NEVER use solvent cement and screw tight outside the gutter.



** NOTE : Fill the groove of coupling with PVC Oil Bottle With Heavy Duty Solvent Cement .

OPTION 2 : Center Mount Method

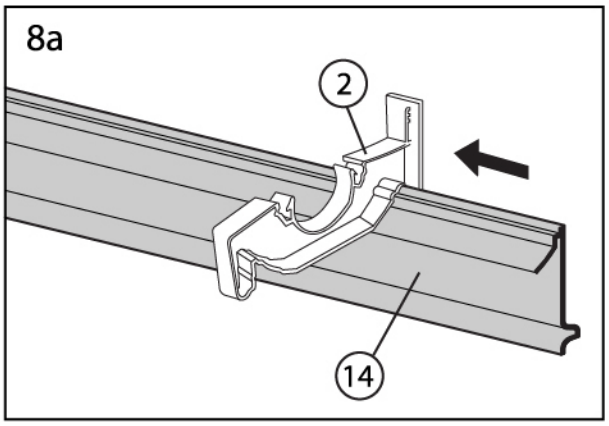


Install the Float Gutter to the center of cooling system.

8. Installing the Top Profile

8a. Pad Guide Installing

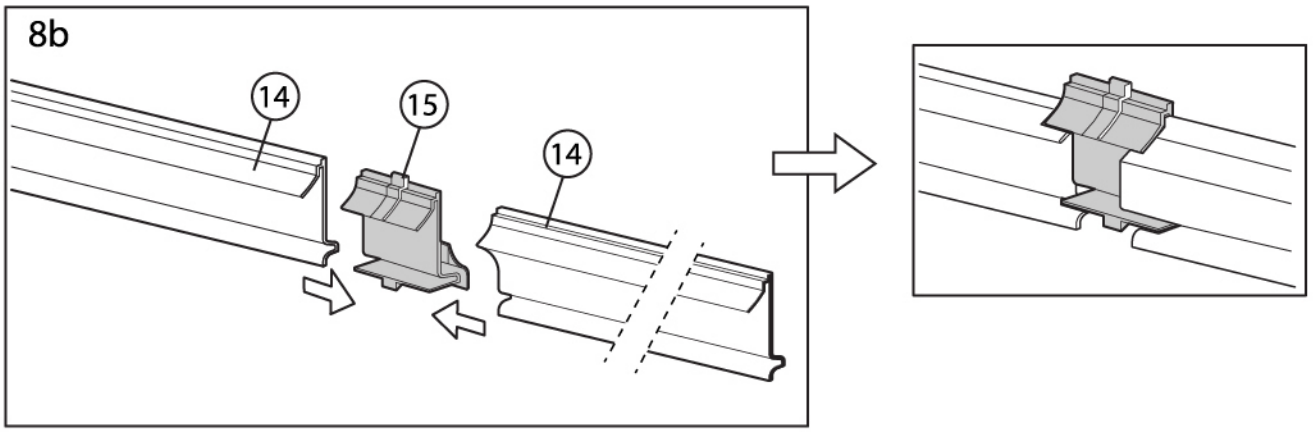
Slide through thee Pad Guide, L3m (14) into the Top Bracket.



Ref #	Description
2	Top Bracket
14	Pad Guide, L3m
15	Pad Guide Coupler
16	Water Distribution Pipe, L3m
17	Top Deflector Cover, L3m
18	Pipe Connector
19	Deflector Coupling

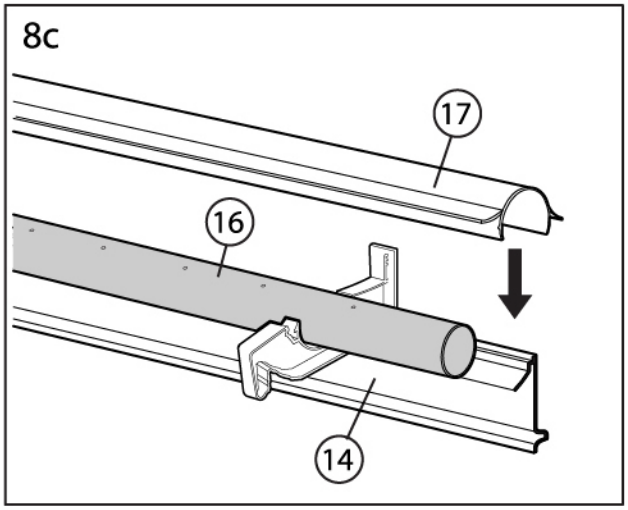
8b. Pad Guide Coupler Installing

Joint the Pad Guide (14) using Pad Guide Coupler (15).

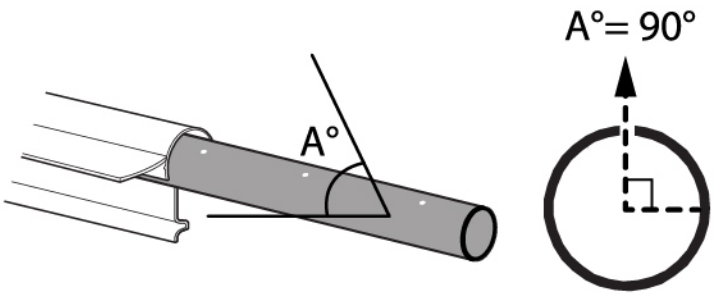


8c. Water distribution Pipe Deflector Installing

Place the Water Distribution Pipe with holes (16) on the top profile and rotate the holes on upward angle. Press the Deflector, L3m (17) down into the pipe.

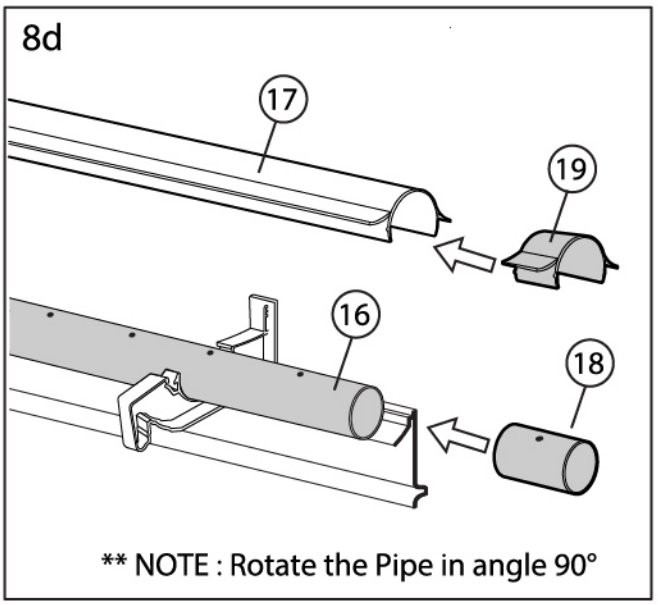


** NOTE : Rotate the Pipe in angle 90°

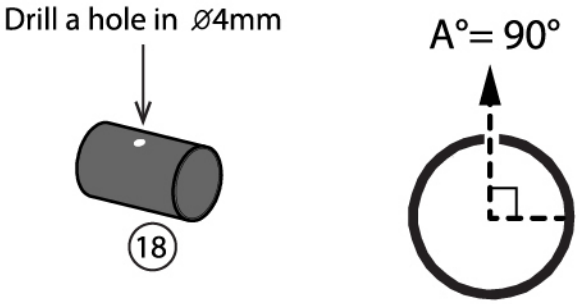


8d. Pipe Connection Assembly

Joint the Deflector (17) to the Deflector Coupling (19).
And connect the Water Distribution Pipe (16) to Pipe Connector (18).



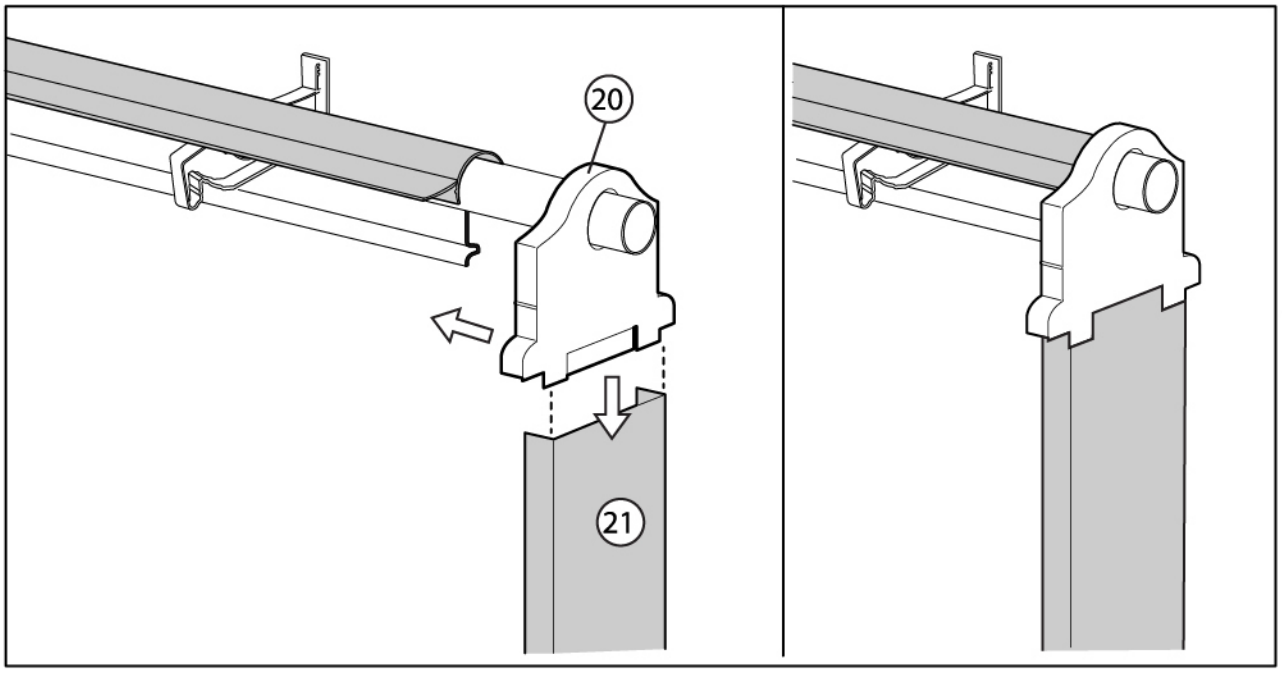
NOTE :
Drill a hole in dia. 4mm on the Pipe Connector (18).



Ref #	Description
16	Water Distribution Pipe, L3m
17	Top Deflector Cover, L3m
18	Pipe Connector
19	Deflector Coupling
20	End Cover for Deflector
21	Pad Cover, L2m

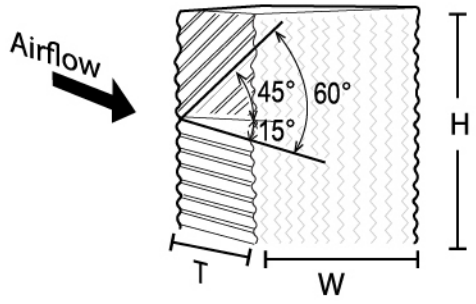
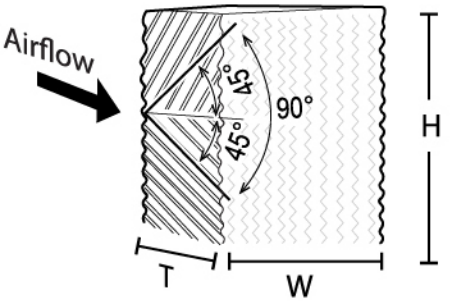
9. Installing the End Cover

Push the End Cover for Deflector (20) on to the Pad Cover, L3m (21), then push the both to the Pad Guide and Deflector.



10. Installing the Cooling Pad

Type of Cool Cell Pad

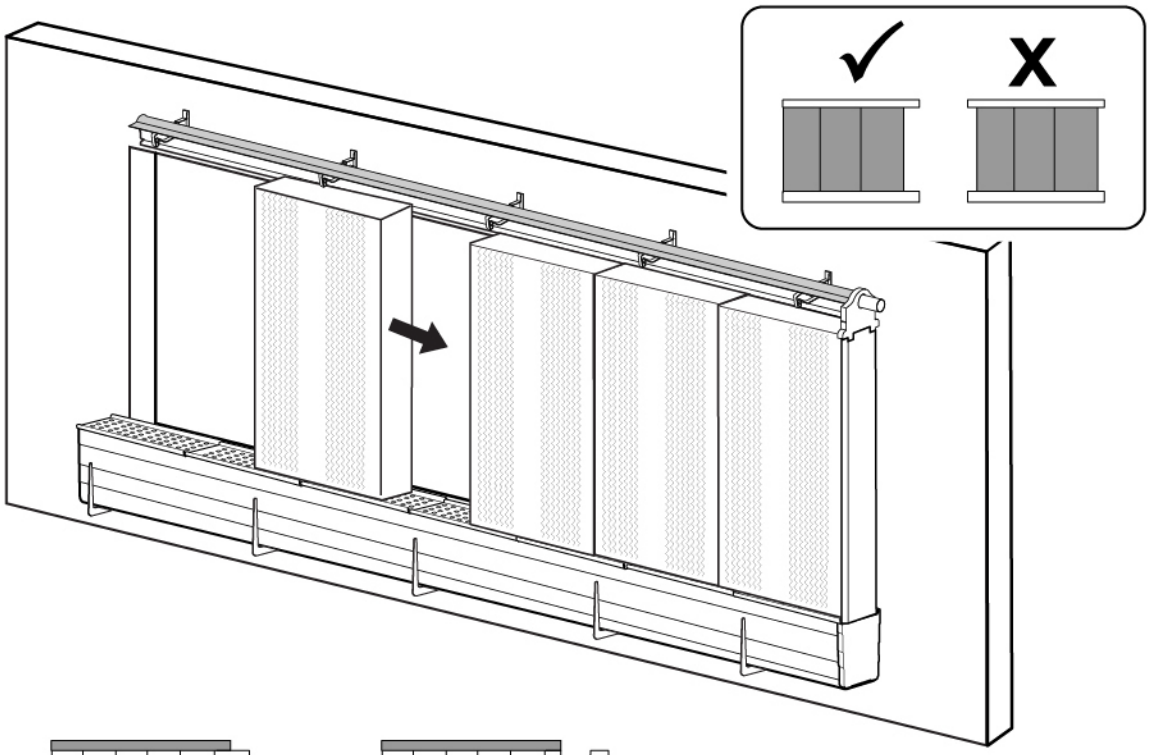
Model : 7060		Model : 7090	
High (H)	1530 & 1830 (5ft & 6ft)	High (H)	1530 & 1830 (5ft & 6ft)
Width (W)	600 mm	Width (W)	600 mm
Thickness (T)	150 mm	Thickness (T)	150 mm
			

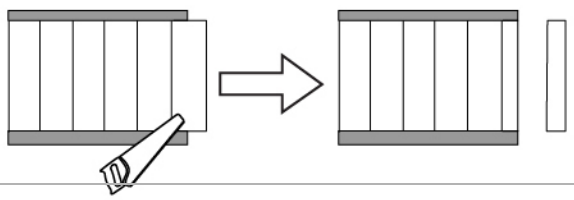
Putting the cool cell in place, take care to position piece tight together, the direction of cool cell is pre-defined. Observe the direction, incorred installation direction can affect the functionality of cool cell.

Shake every pad and distributor before inserting in order to remove dust.

Insert the cooling pad as indicated in below. Ensure that all pads are moved tightly together once inserted.

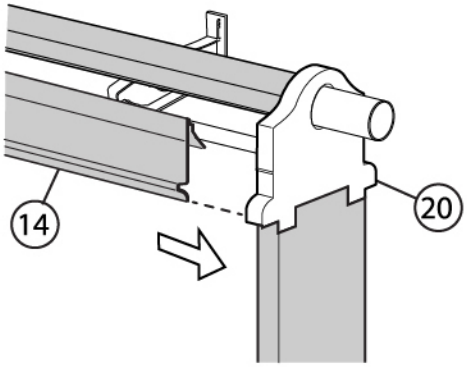
Ensure that the vertical alignment between the cooling pad and the gutters is in line as indicated.





The last pad and distributor may need to be trimmed. Use a saw if trimming is required.

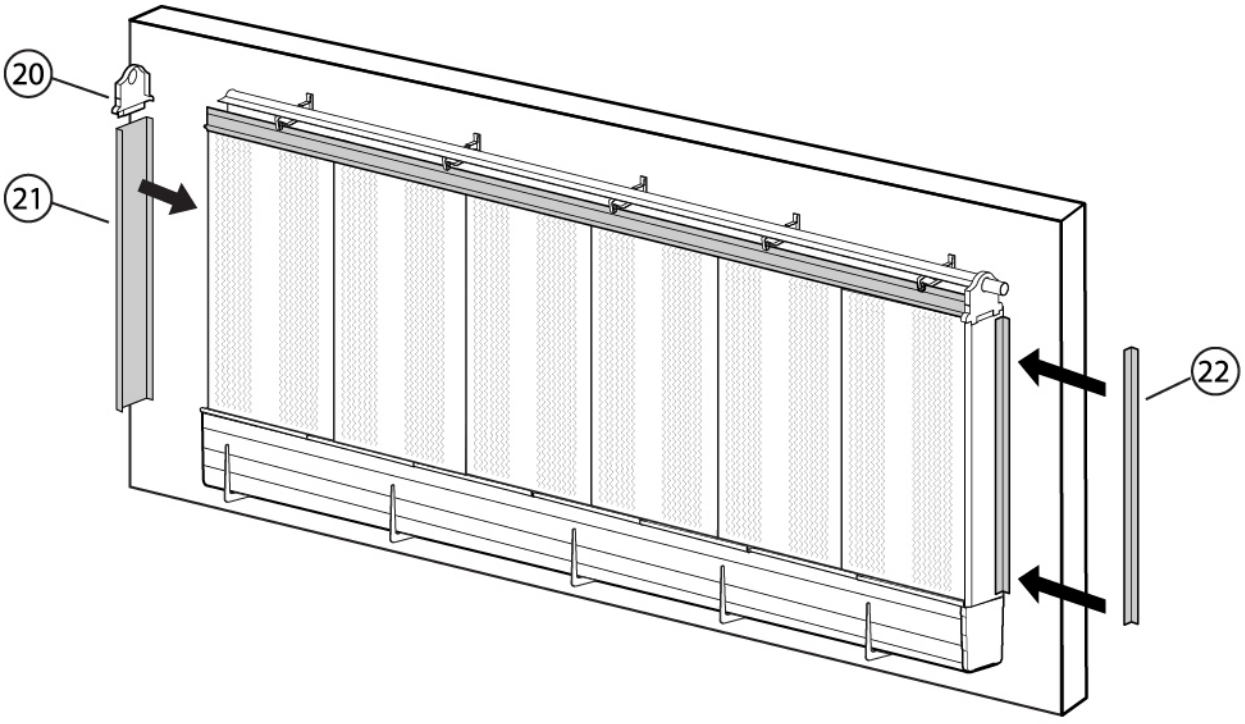
Attach the another Pad Guide (14) to the End Cover for Deflector (20).



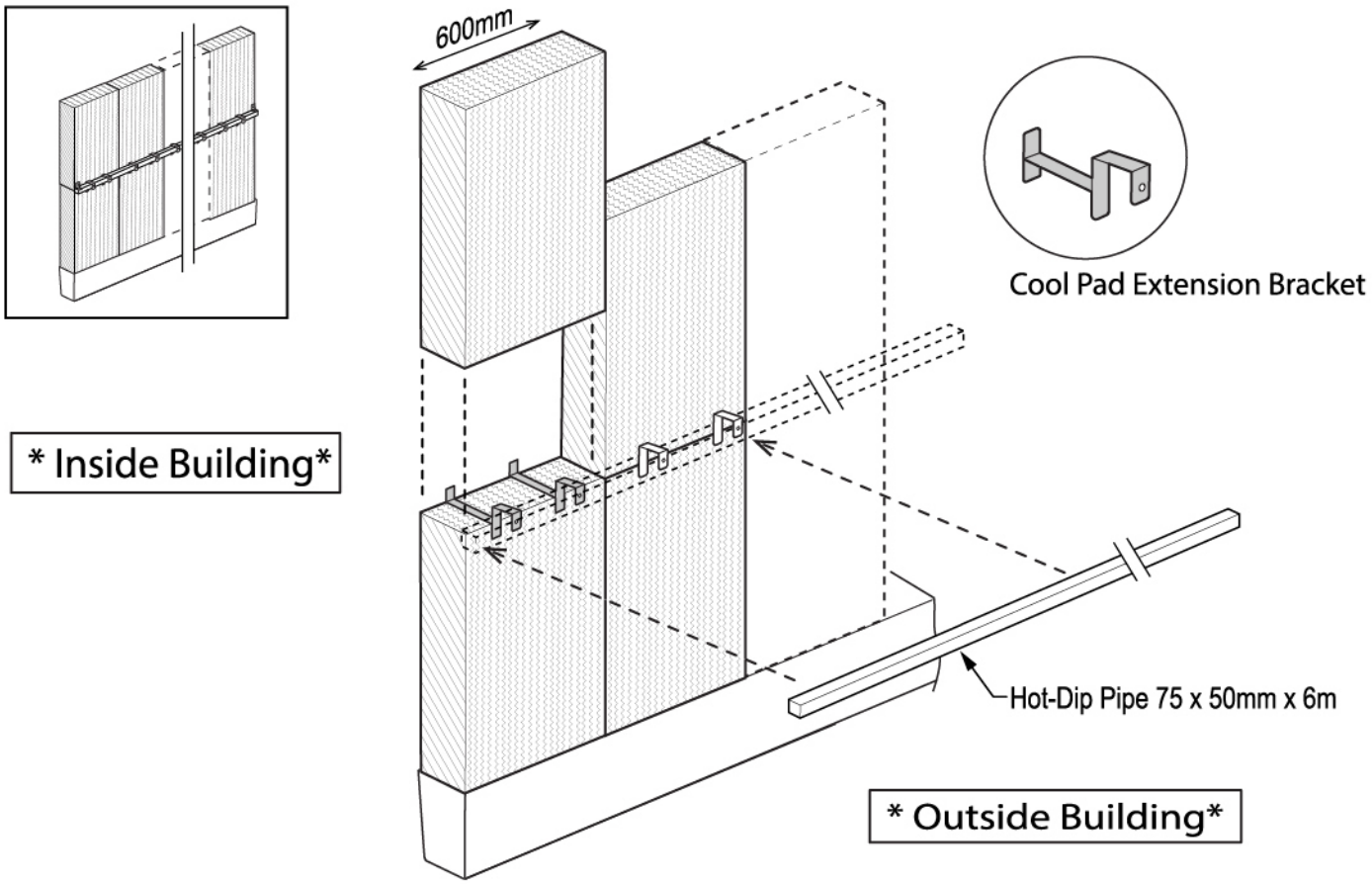
Ref #	Description
14	Pad Guide, L3m
20	End Cover for Deflector
21	Pad Cover, L2m
22	L-Capping (Lily White) 50x50x0.48x L8ft

---

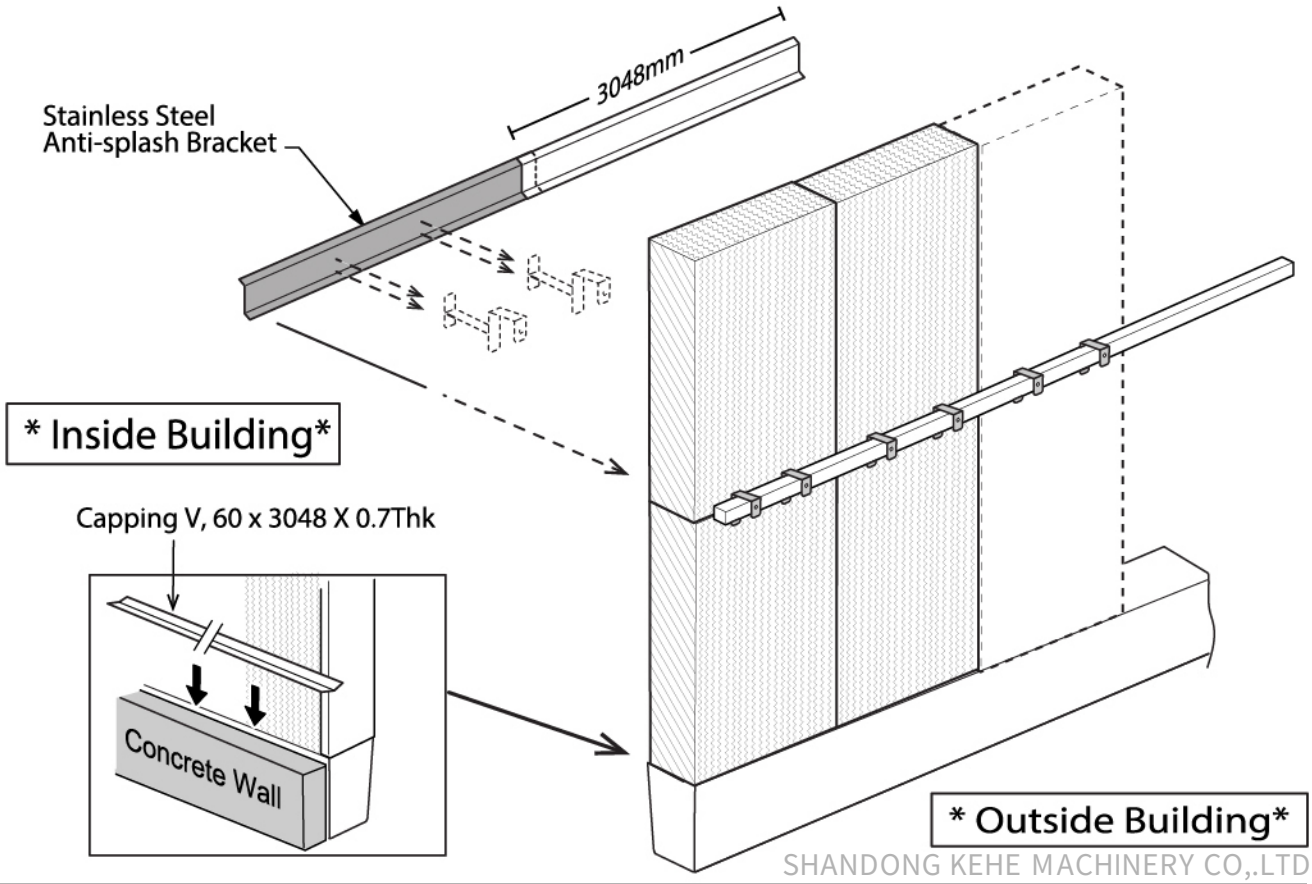
1. Push the End Cover for Deflector (20) on to the Pad cover (21), then push the both cover to cool cell.
2. Cut the L-Capping 50x50 (22) to require length, and cover the gap between Pad Cover (21)and the wall. Secure by hardware provided.



11. Cool Pad Extension Bracket Assembly and Water Supply

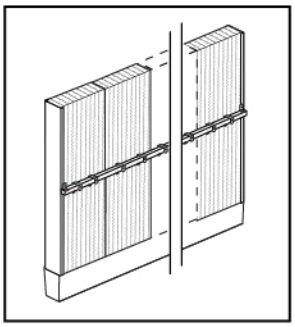


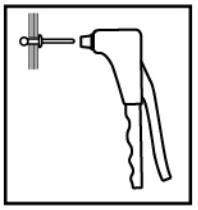
11a. Anti-Splash Bracket and Connector Assembly

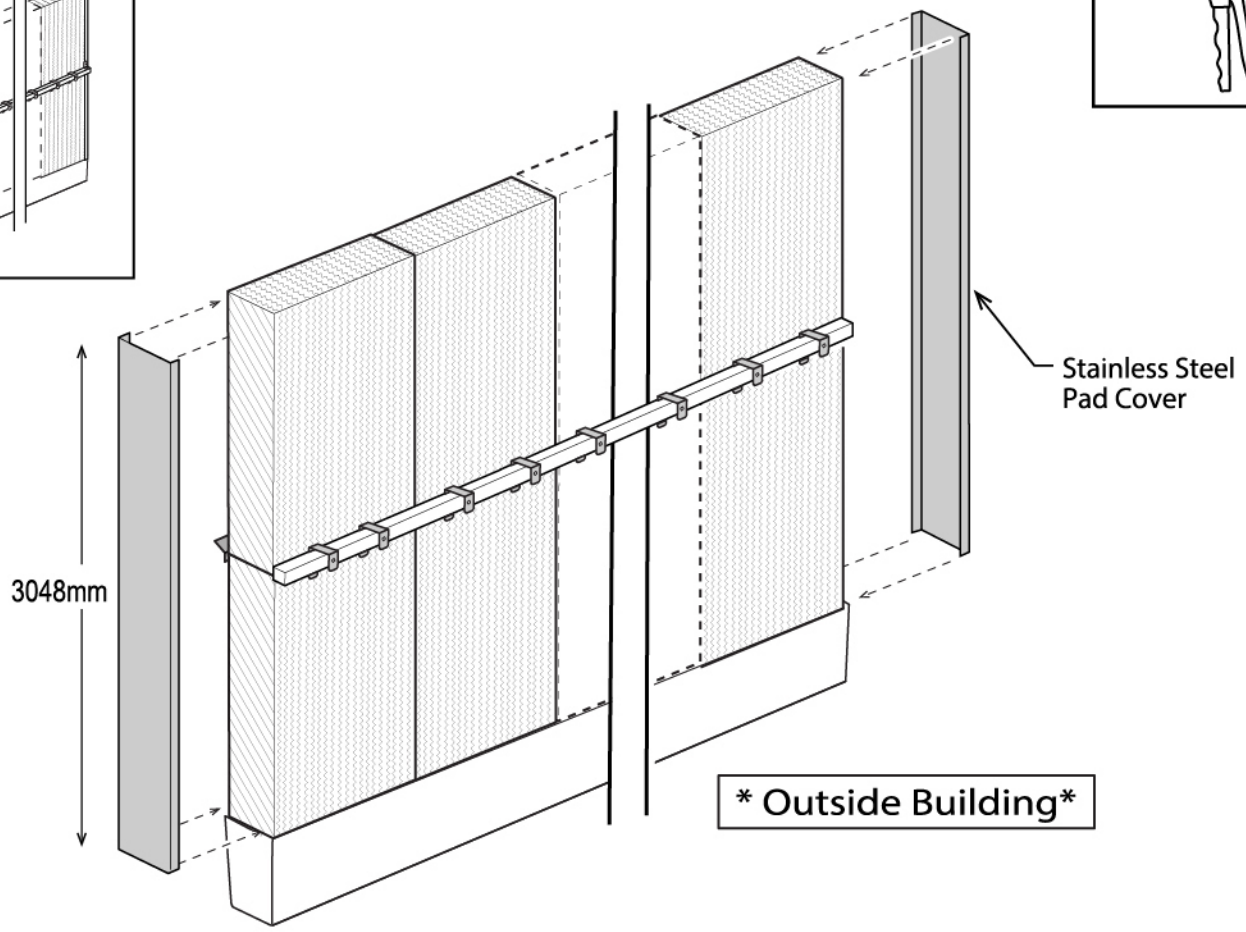


SHANDONG KEHE MACHINERY CO,.LTD

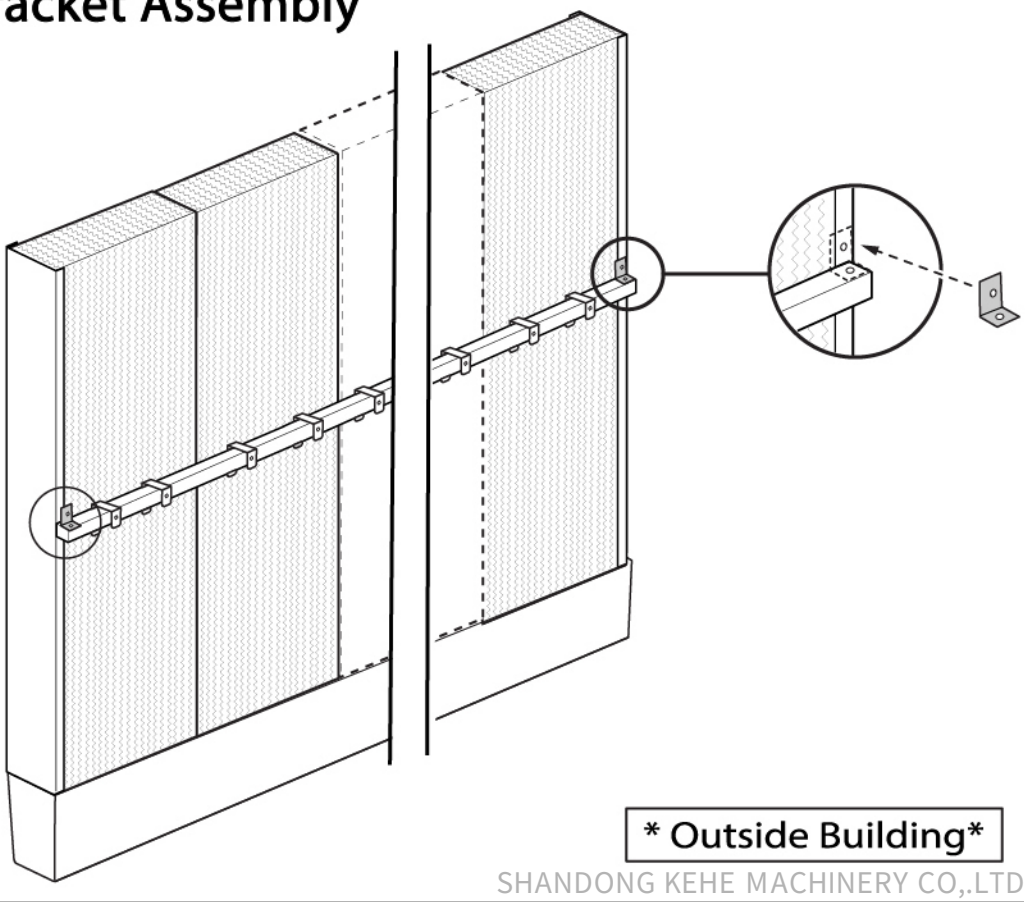
11b. Stainless Steel Pad Cover Assembly





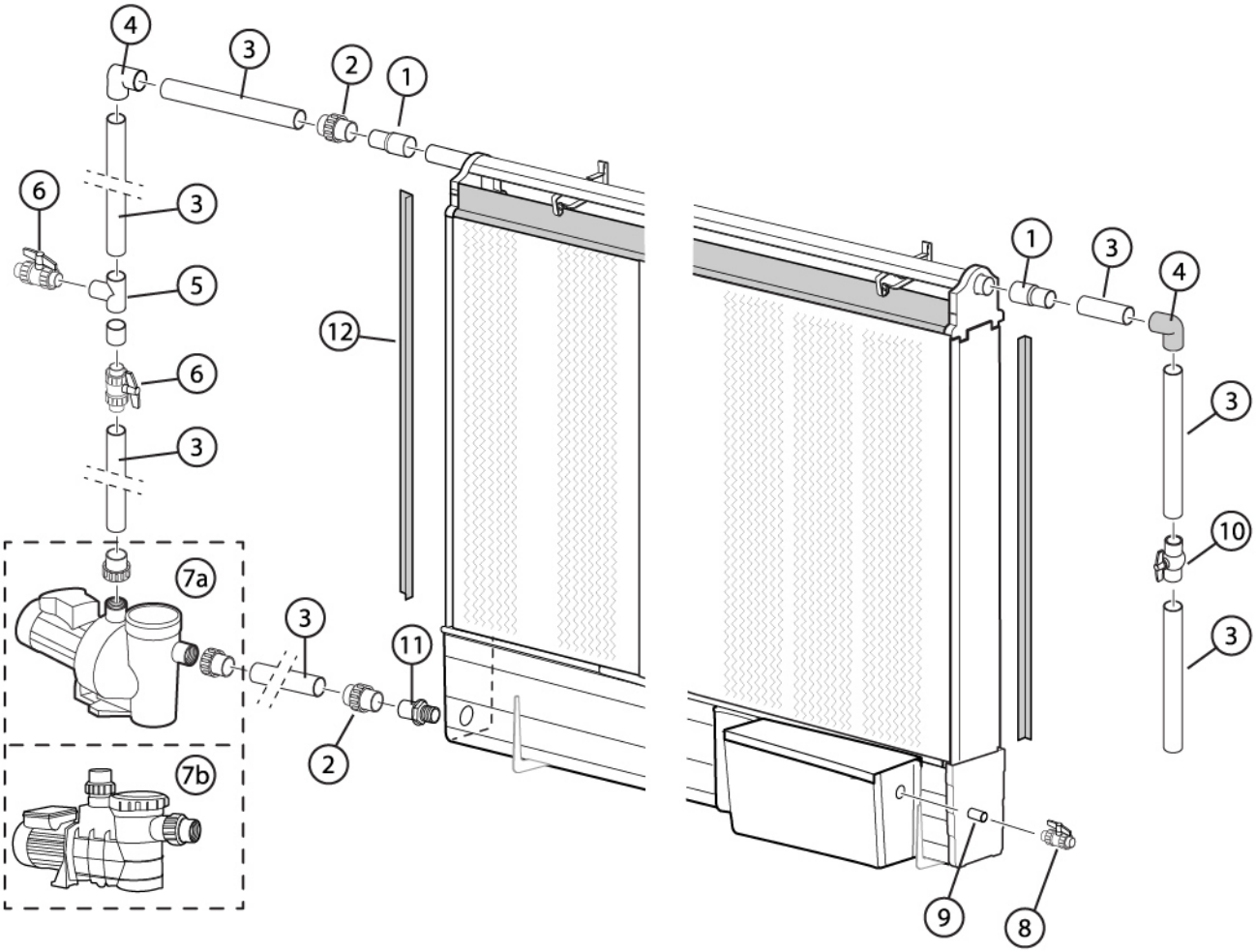


11c. Support Bracket Assembly



12. Part List Distribution

Note : The plumbing illustration just for recommendation only. Please source locally when the PVC pipes are no adequate of installation.



Ref #	Description	Packing List
1	60-48mm (2" to 1-½") PVC Reduction	Box 2
2	48mm PVC Coupling	Box 2
3	PVC Pipe 1-1/2" (48mm)	Box 2 (L2m)
4	48mm (1-1/2") PVC Elbow 90°	Box 2
5	48mm (1-½") PVC Equal Tee	Box 2
6	48mm (1-½") PVC Ball Valve with Coupling	Box 2
7a	Water Pump , 0.75Kw/1.0Hp/220V/50Hz	
7b	Water Pump HLS-750, 0.75Kw, 415V 3P 50Hz	
8	25mm (¾") PVC Ball Valve with Coupling	Box 2
9	25mm (¾") Pipe	Box 2
10	48mm (1-½") PVC Ball Valve	Box 2
11	PVC Tank Connector 1-1/2"	
12	Capping L 100mm x 2438mm (8ft) x 0.48thk	

13. Water Pump Specification

Pump Technical Data

Power Output	0.37Kw
Weight	8.1Kgs
Capaity	180L /min
Water pump head	9m
Electricity	220V/415V/380V 50Hz/60Hz

Power Output	0.75Kw
Weight	9Kgs
Capaity	240L /min
Water pump head	11m
Electricity	220V/415V/380V 50Hz/60Hz

Power Output	1.1Kw
Weight	14.6Kgs
Capaity	360L /min
Water pump head	10m
Electricity	220V/415V/380V 50Hz/60Hz

Power Output	1.5Kw
Weight	14.8Kgs
Capaity	400L /min
Water pump head	12m
Electricity	220V/415V/380V 50Hz/60Hz

14. Maintenance

To maintain your cooling system in top condition, the recommendations given in Chart A should be treated as minimums. More frequent maintenance may be required at initial system start-up, in certain climates, and in areas with hard water conditions.

MINIMUM MAINTENANCE SCHEDULE, Chart A

Maintenance Operation	SCHEDULE: Weekly	Monthly	Yearly
Clean the foot valve or the pump filter screen		X	
Flush in-line strainer	X		
Clean return water filter screen	X		
Check for dry streaks on pad material	X		
Clean debris from face of pad material	X		
Clean cooling control		X	
Clean cooling panel distribution pipe holes		X	
Clean cooling panel collection trough			X
Drain and clean sump tank		X	
Clean float valve		X	

Maintenance Checklist

Follow the guide lines below to prevent early pad failure and to get the maximum life from your cool pads.

1. Reduce the number of on-off cycles. Do Not use 10 minute cycle timers on pumps.
2. Shade the pads and sump to minimize algae growth.
3. Dry the pads out completely each night to kill algae. A 24 hour timer may be installed to shut pumps off at 10 pm and allow them to come on after 9 am.
4. Bleed off some water continuously to prevent concentration of minerals and dissolved chemicals in sump. Start with 1 gallon/hour for each linear foot of cooling system and adjust as necessary.
5. Drain and disinfect the water distribution system every three months to minimize algae, fungus and mineral build-up.
6. Periodically check for leaks in water distribution system.
7. Avoid harmful contaminants such as dust, fumes, fertilizers, harsh cleaners and water treatment chemicals.
8. Complete the items on the Maintenance Schedule as required.



WARNING: DO NOT add chemicals or commercial water treatments to the sump or supply water.